

## 1. Identification

**Other means of identification** None known.

**Product identifier** **HYDROXYLAMINE SULFATE**

**Other means of identification**

**CAS number** 10039-54-0

**Recommended use** ALL PROPER AND LEGAL PURPOSES

**Recommended restrictions** None known.

**Manufacturer/Importer/Supplier/Distributor information**

**Manufacturer**

**Company name** Brenntag Mid-South, Inc.

**Address** 1405 Highway 136, West  
Henderson, KY 42420

**Telephone** 270-830-1222

**E-mail** Not available.

**Emergency phone number** 800-424-9300 CHEMTREC

## 2. Hazard(s) identification

**Physical hazards** Corrosive to metals Category 1

**Health hazards** Acute toxicity, oral Category 4

Acute toxicity, dermal Category 4

Skin corrosion/irritation Category 1

Serious eye damage/eye irritation Category 1

Sensitization, skin Category 1

Carcinogenicity Category 2

Specific target organ toxicity, repeated exposure Category 2

**Environmental hazards** Not classified.

**OSHA defined hazards** Not classified.

**Label elements**



**Signal word** Danger

**Hazard statement** May be corrosive to metals. Harmful if swallowed. Harmful in contact with skin. Causes severe skin burns and eye damage. May cause an allergic skin reaction. Causes serious eye damage. Suspected of causing cancer. May cause damage to organs through prolonged or repeated exposure.

### Precautionary statement

#### Prevention

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Keep only in original container. Do not breathe dust/fume/gas/mist/vapors/spray. Wash thoroughly after handling. Do not eat, drink or smoke when using this product. Contaminated work clothing must not be allowed out of the workplace. Wear protective gloves/protective clothing/eye protection/face protection.

#### Response

If swallowed: Rinse mouth. Do NOT induce vomiting. If on skin (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower. If inhaled: Remove person to fresh air and keep comfortable for breathing. If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a poison center/doctor. If skin irritation or rash occurs: Get medical advice/attention. Absorb spillage to prevent material damage.

#### Storage

Store locked up. Store in corrosive resistant container with a resistant inner liner.

|                                                  |                                                                                                     |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Disposal</b>                                  | Dispose of contents/container in accordance with local/regional/national/international regulations. |
| <b>Hazard(s) not otherwise classified (HNOC)</b> | None known.                                                                                         |
| <b>Supplemental information</b>                  | None.                                                                                               |

### 3. Composition/information on ingredients

#### Substances

| Chemical name                  | Common name and synonyms | CAS number | %   |
|--------------------------------|--------------------------|------------|-----|
| Bis(hydroxylammonium) sulphate |                          | 10039-54-0 | 100 |

### 4. First-aid measures

|                                                                               |                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                                                             | Move to fresh air. Call a physician if symptoms develop or persist.                                                                                                                                                                                                                                                                                |
| <b>Skin contact</b>                                                           | Remove contaminated clothing immediately and wash skin with soap and water. Call a physician or poison control center immediately. Chemical burns must be treated by a physician. Wash contaminated clothing before reuse.                                                                                                                         |
| <b>Eye contact</b>                                                            | Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Call a physician or poison control center immediately.                                                                                                                                                    |
| <b>Ingestion</b>                                                              | Call a physician or poison control center immediately. Rinse mouth. Do not induce vomiting. If vomiting occurs, keep head low so that stomach content doesn't get into the lungs.                                                                                                                                                                  |
| <b>Most important symptoms/effects, acute and delayed</b>                     | Burning pain and severe corrosive skin damage. Causes serious eye damage. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. Permanent eye damage including blindness could result. Prolonged exposure may cause chronic effects.                                                                                      |
| <b>Indication of immediate medical attention and special treatment needed</b> | Provide general supportive measures and treat symptomatically. Chemical burns: Flush with water immediately. While flushing, remove clothes which do not adhere to affected area. Call an ambulance. Continue flushing during transport to hospital. Keep victim warm. Keep victim under observation. Symptoms may be delayed.                     |
| <b>General information</b>                                                    | IF exposed or concerned: Get medical advice/attention. If you feel unwell, seek medical advice (show the label where possible). Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves. Show this safety data sheet to the doctor in attendance. Wash contaminated clothing before reuse. |

### 5. Fire-fighting measures

|                                                                      |                                                                                               |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Suitable extinguishing media</b>                                  | Alcohol resistant foam. Powder. Carbon dioxide (CO <sub>2</sub> ).                            |
| <b>Unsuitable extinguishing media</b>                                | Do not use water jet as an extinguisher, as this will spread the fire.                        |
| <b>Specific hazards arising from the chemical</b>                    | During fire, gases hazardous to health may be formed.                                         |
| <b>Special protective equipment and precautions for firefighters</b> | Self-contained breathing apparatus and full protective clothing must be worn in case of fire. |
| <b>Fire fighting equipment/instructions</b>                          | Move containers from fire area if you can do so without risk.                                 |
| <b>Specific methods</b>                                              | Use standard firefighting procedures and consider the hazards of other involved materials.    |

### 6. Accidental release measures

|                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions, protective equipment and emergency procedures</b> | Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Wear appropriate protective equipment and clothing during clean-up. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ensure adequate ventilation. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.                                                                                                                                                                                                                                                 |
| <b>Methods and materials for containment and cleaning up</b>               | Prevent entry into waterways, sewer, basements or confined areas.<br><br>Large Spills: Stop the flow of material, if this is without risk. Dike the spilled material, where this is possible. Absorb spillage to prevent material damage. Absorb in vermiculite, dry sand or earth and place into containers. Following product recovery, flush area with water.<br><br>Small Spills: Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to remove residual contamination.<br><br>Never return spills to original containers for re-use. Put material in suitable, covered, labeled containers. For waste disposal, see section 13 of the SDS. |
| <b>Environmental precautions</b>                                           | Avoid discharge into drains, water courses or onto the ground.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## 7. Handling and storage

### Precautions for safe handling

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Do not get in eyes, on skin, or on clothing. Do not taste or swallow. Avoid breathing dust/fume/gas/mist/vapors/spray. When using, do not eat, drink or smoke. Should be handled in closed systems, if possible. Provide adequate ventilation. Wear appropriate personal protective equipment. Wash hands thoroughly after handling. Wash contaminated clothing before reuse. Observe good industrial hygiene practices.

### Conditions for safe storage, including any incompatibilities

Store locked up. Store in a cool, dry place out of direct sunlight. Store in corrosive resistant container with a resistant inner liner. Store in tightly closed container. Keep only in the original container. Store away from incompatible materials (see Section 10 of the SDS).

## 8. Exposure controls/personal protection

### Occupational exposure limits

No exposure limits noted for ingredient(s).

### Biological limit values

No biological exposure limits noted for the ingredient(s).

### Appropriate engineering controls

Good general ventilation should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. Eye wash facilities and emergency shower must be available when handling this product.

### Individual protection measures, such as personal protective equipment

The following are recommendations for Personnel Protective Equipment (PPE). The employer/user of this product must perform a Hazard Assessment of the workplace according to OSHA regulations 29 CFR 1910.132 to determine the appropriate PPE for use while performing any task involving potential exposure to this product.

#### Eye/face protection

Wear safety glasses with side shields (or goggles) and a face shield.

#### Skin protection

##### Hand protection

Wear appropriate chemical resistant gloves.

##### Other

Wear appropriate chemical resistant clothing. Use of an impervious apron is recommended.

#### Respiratory protection

In case of insufficient ventilation, wear suitable respiratory equipment.

#### Thermal hazards

Wear appropriate thermal protective clothing, when necessary.

### General hygiene considerations

Observe any medical surveillance requirements. Keep away from food and drink. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants. Contaminated work clothing should not be allowed out of the workplace.

## 9. Physical and chemical properties

### Appearance

#### Physical state

Solid.

#### Form

Solid.

#### Color

WHITE

### Odor

SLIGHT CHARACTERISTIC

### Odor threshold

Not available.

### pH

Not available.

### Melting point/freezing point

338 °F (170 °C)

### Initial boiling point and boiling range

Not available.

### Flash point

Not available.

### Evaporation rate

Not available.

### Flammability (solid, gas)

Not available.

### Upper/lower flammability or explosive limits

#### Flammability limit - lower (%)

Not available.

#### Flammability limit - upper (%)

Not available.

#### Explosive limit - lower (%)

Not available.

#### Explosive limit - upper (%)

Not available.

### Vapor pressure

Not available.

|                                                |                           |
|------------------------------------------------|---------------------------|
| <b>Vapor density</b>                           | Not available.            |
| <b>Relative density</b>                        | Not available.            |
| <b>Solubility(ies)</b>                         |                           |
| <b>Solubility (water)</b>                      | Not available.            |
| <b>Partition coefficient (n-octanol/water)</b> | Not available.            |
| <b>Auto-ignition temperature</b>               | Not available.            |
| <b>Decomposition temperature</b>               | Not available.            |
| <b>Viscosity</b>                               | Not available.            |
| <b>Other information</b>                       |                           |
| <b>Density</b>                                 | 1.00 lbs/gal<br>0.12 g/ml |
| <b>Explosive properties</b>                    | Not explosive.            |
| <b>Molecular formula</b>                       | H3NO.1/2H2O4S             |
| <b>Oxidizing properties</b>                    | Not oxidizing.            |
| <b>Specific gravity</b>                        | 0.12                      |

## 10. Stability and reactivity

|                                           |                                                |
|-------------------------------------------|------------------------------------------------|
| <b>Reactivity</b>                         | May be corrosive to metals.                    |
| <b>Chemical stability</b>                 | Material is stable under normal conditions.    |
| <b>Possibility of hazardous reactions</b> | Hazardous polymerization does not occur.       |
| <b>Conditions to avoid</b>                | Contact with incompatible materials.           |
| <b>Incompatible materials</b>             | Strong oxidizing agents. Metals.               |
| <b>Hazardous decomposition products</b>   | No hazardous decomposition products are known. |

## 11. Toxicological information

### Information on likely routes of exposure

|                     |                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | May cause irritation to the respiratory system.                                              |
| <b>Skin contact</b> | Causes severe skin burns. Harmful in contact with skin. May cause an allergic skin reaction. |
| <b>Eye contact</b>  | Causes serious eye damage.                                                                   |
| <b>Ingestion</b>    | Causes digestive tract burns. Harmful if swallowed.                                          |

|                                                                                     |                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Symptoms related to the physical, chemical and toxicological characteristics</b> | Burning pain and severe corrosive skin damage. Causes serious eye damage. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. Permanent eye damage including blindness could result. |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Information on toxicological effects

|                                          |                                                     |
|------------------------------------------|-----------------------------------------------------|
| <b>Acute toxicity</b>                    | Harmful in contact with skin. Harmful if swallowed. |
| <b>Skin corrosion/irritation</b>         | Causes severe skin burns and eye damage.            |
| <b>Serious eye damage/eye irritation</b> | Causes serious eye damage.                          |

### Respiratory or skin sensitization

|                                  |                                                                             |
|----------------------------------|-----------------------------------------------------------------------------|
| <b>Respiratory sensitization</b> | Due to partial or complete lack of data the classification is not possible. |
| <b>Skin sensitization</b>        | May cause an allergic skin reaction.                                        |

|                               |                                                                             |
|-------------------------------|-----------------------------------------------------------------------------|
| <b>Germ cell mutagenicity</b> | Due to partial or complete lack of data the classification is not possible. |
|-------------------------------|-----------------------------------------------------------------------------|

|                        |                              |
|------------------------|------------------------------|
| <b>Carcinogenicity</b> | Suspected of causing cancer. |
|------------------------|------------------------------|

#### IARC Monographs. Overall Evaluation of Carcinogenicity

Not listed.

#### OSHA Specifically Regulated Substances (29 CFR 1910.1001-1053)

Not listed.

#### US. National Toxicology Program (NTP) Report on Carcinogens

Not listed.

|                              |                                                                             |
|------------------------------|-----------------------------------------------------------------------------|
| <b>Reproductive toxicity</b> | Due to partial or complete lack of data the classification is not possible. |
|------------------------------|-----------------------------------------------------------------------------|

|                                                           |                                                                             |
|-----------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Specific target organ toxicity - single exposure</b>   | Due to partial or complete lack of data the classification is not possible. |
| <b>Specific target organ toxicity - repeated exposure</b> | May cause damage to organs through prolonged or repeated exposure.          |
| <b>Aspiration hazard</b>                                  | Due to partial or complete lack of data the classification is not possible. |
| <b>Chronic effects</b>                                    | May cause damage to organs through prolonged or repeated exposure.          |

## 12. Ecological information

|                                      |                                                                                                                                                                                                    |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ecotoxicity</b>                   | The product is not classified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment. |
| <b>Persistence and degradability</b> | No data is available on the degradability of this product.                                                                                                                                         |
| <b>Bioaccumulative potential</b>     | No data available.                                                                                                                                                                                 |
| <b>Mobility in soil</b>              | No data available.                                                                                                                                                                                 |
| <b>Other adverse effects</b>         | No other adverse environmental effects (e.g. ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected from this component.            |

## 13. Disposal considerations

|                                              |                                                                                                                                                                                                                                                                          |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Disposal instructions</b>                 | Collect and reclaim or dispose in sealed containers at licensed waste disposal site. Incinerate the material under controlled conditions in an approved incinerator. Dispose of contents/container in accordance with local/regional/national/international regulations. |
| <b>Local disposal regulations</b>            | Dispose in accordance with all applicable regulations.                                                                                                                                                                                                                   |
| <b>Hazardous waste code</b>                  | D002: Waste Corrosive material [pH ≤2 or ≥12.5, or corrosive to steel]<br>The waste code should be assigned in discussion between the user, the producer and the waste disposal company.                                                                                 |
| <b>Waste from residues / unused products</b> | Dispose of in accordance with local regulations. Empty containers or liners may retain some product residues. This material and its container must be disposed of in a safe manner (see: Disposal instructions).                                                         |
| <b>Contaminated packaging</b>                | Since emptied containers may retain product residue, follow label warnings even after container is emptied. Empty containers should be taken to an approved waste handling site for recycling or disposal.                                                               |

## 14. Transport information

|                                                                                                                                                  |                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>DOT</b>                                                                                                                                       |                                                                         |
| <b>UN number</b>                                                                                                                                 | UN2865                                                                  |
| <b>UN proper shipping name</b>                                                                                                                   | HYDROXYLAMINE SULFATE                                                   |
| <b>Transport hazard class(es)</b>                                                                                                                |                                                                         |
| <b>Class</b>                                                                                                                                     | 8                                                                       |
| <b>Subsidiary risk</b>                                                                                                                           | -                                                                       |
| <b>Packing group</b>                                                                                                                             | III                                                                     |
| <b>Special precautions for user</b>                                                                                                              | Read safety instructions, SDS and emergency procedures before handling. |
| <b>ERG number</b>                                                                                                                                | 154                                                                     |
| Transport information on packaging may be different from that listed. Transportation information on packaging may be different from that listed. |                                                                         |

### IATA

|                                     |                                                                         |
|-------------------------------------|-------------------------------------------------------------------------|
| <b>UN number</b>                    | UN2865                                                                  |
| <b>UN proper shipping name</b>      | HYDROXYLAMINE SULFATE                                                   |
| <b>Transport hazard class(es)</b>   |                                                                         |
| <b>Class</b>                        | 8                                                                       |
| <b>Subsidiary risk</b>              | -                                                                       |
| <b>Packing group</b>                | III                                                                     |
| <b>Environmental hazards</b>        | No.                                                                     |
| <b>ERG Code</b>                     | 154                                                                     |
| <b>Special precautions for user</b> | Read safety instructions, SDS and emergency procedures before handling. |

### IMDG

|                                   |                        |
|-----------------------------------|------------------------|
| <b>UN number</b>                  | UN2865                 |
| <b>UN proper shipping name</b>    | HYDROXYLAMINE SULPHATE |
| <b>Transport hazard class(es)</b> |                        |
| <b>Class</b>                      | 8                      |
| <b>Subsidiary risk</b>            | -                      |
| <b>Packing group</b>              | III                    |

**Environmental hazards****Marine pollutant**

No.

**EmS**

F-A, S-B

**Special precautions for user** Read safety instructions, SDS and emergency procedures before handling.**DOT****IATA; IMDG****15. Regulatory information****US federal regulations**

This product is a "Hazardous Chemical" as defined by the OSHA Hazard Communication Standard, 29 CFR 1910.1200.

**Toxic Substances Control Act (TSCA)****TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D)**

Not regulated.

**CERCLA Hazardous Substance List (40 CFR 302.4)**

Not listed.

**SARA 304 Emergency release notification**

Not regulated.

**OSHA Specifically Regulated Substances (29 CFR 1910.1001-1053)**

Not listed.

**Superfund Amendments and Reauthorization Act of 1986 (SARA)****SARA 302 Extremely hazardous substance**

Not listed.

**SARA 311/312 Hazardous chemical**

Yes

**Classified hazard categories**

Corrosive to metal  
Acute toxicity (any route of exposure)  
Skin corrosion or irritation  
Serious eye damage or eye irritation  
Respiratory or skin sensitization  
Carcinogenicity  
Specific target organ toxicity (single or repeated exposure)

**SARA 313 (TRI reporting)**

Not regulated.

**Other federal regulations****Clean Air Act (CAA) Section 112 Hazardous Air Pollutants (HAPs) List**

Not regulated.

**Clean Air Act (CAA) Section 112(r) Accidental Release Prevention (40 CFR 68.130)**

Not regulated.

**Safe Drinking Water Act (SDWA)** Not regulated.

#### US state regulations

##### California Proposition 65

California Safe Drinking Water and Toxic Enforcement Act of 1986 (Proposition 65): This material is not known to contain any chemicals currently listed as carcinogens or reproductive toxins. For more information go to [www.P65Warnings.ca.gov](http://www.P65Warnings.ca.gov).

#### International Inventories

| Country(s) or region        | Inventory name                                                         | On inventory (yes/no)* |
|-----------------------------|------------------------------------------------------------------------|------------------------|
| Australia                   | Australian Inventory of Chemical Substances (AICS)                     | Yes                    |
| Canada                      | Domestic Substances List (DSL)                                         | Yes                    |
| Canada                      | Non-Domestic Substances List (NDSL)                                    | No                     |
| China                       | Inventory of Existing Chemical Substances in China (IECSC)             | Yes                    |
| Europe                      | European Inventory of Existing Commercial Chemical Substances (EINECS) | Yes                    |
| Europe                      | European List of Notified Chemical Substances (ELINCS)                 | No                     |
| Japan                       | Inventory of Existing and New Chemical Substances (ENCS)               | Yes                    |
| Korea                       | Existing Chemicals List (ECL)                                          | Yes                    |
| New Zealand                 | New Zealand Inventory                                                  | Yes                    |
| Philippines                 | Philippine Inventory of Chemicals and Chemical Substances (PICCS)      | Yes                    |
| Taiwan                      | Taiwan Chemical Substance Inventory (TCSI)                             | Yes                    |
| United States & Puerto Rico | Toxic Substances Control Act (TSCA) Inventory                          | Yes                    |

\*A "Yes" indicates that all components of this product comply with the inventory requirements administered by the governing country(s)

A "No" indicates that one or more components of the product are not listed or exempt from listing on the inventory administered by the governing country(s).

#### 16. Other information, including date of preparation or last revision

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Issue date</b>           | 05-22-2015                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Revision date</b>        | 06-24-2021                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Version #</b>            | 04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>HMIS® ratings</b>        | Health: 3*<br>Flammability: 0<br>Physical hazard: 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>NFPA ratings</b>         | Health: 3<br>Flammability: 0<br>Instability: 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Disclaimer</b>           | While Brenntag believes the information contained herein to be accurate, Brenntag makes no representation or warranty, express or implied, regarding, and assumes no liability for, the accuracy or completeness of the information. The Buyer assumes all responsibility for handling, using and/or reselling the Product in accordance with applicable federal, state, and local law. This SDS shall not in any way limit or preclude the operation and effect of any of the provisions of Brenntag's terms and conditions of sale. |
| <b>Revision information</b> | This document has undergone significant changes and should be reviewed in its entirety.                                                                                                                                                                                                                                                                                                                                                                                                                                               |